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IoT Challenge 1 - Optimizing Sink Position in a Wireless Sensor Net- work

TESI DI LAUREA MAGISTRALE IN
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Author: **Hong CHEN**

Student ID: [REDACTED]

Advisor: Prof. Matteo Cesana

Co-advisors: Alessandro Redondi

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1 | Optimizing Sink Position in a Wireless Sensor Network

10 Sensors with fixed coordinates (See Figure 1.1)

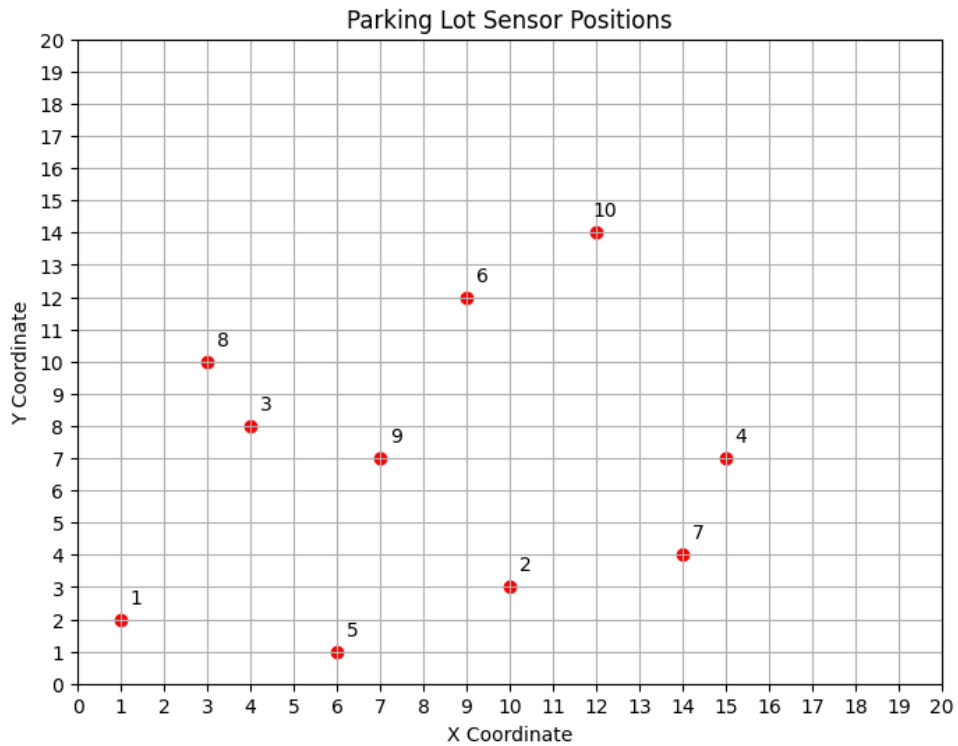


Figure 1.1: The position of 10 sensors

- Initial energy per sensor: 5 mJ
- Packet size: 2000 bits
- Energy for TX/RX circuitry: 50 nJ/bit
- Energy for transmission:

$$E_{tx}(d) = k \cdot d^2 \quad \text{nJ/bit} \quad (\text{where } k = 1 \quad \text{bit/m}^2)$$

The distance between sensor and the sink at is calculated using the Euclidean distance formula:

$$d_i = \sqrt{(x_i - x_s)^2 + (y_i - y_s)^2}$$

The total energy consumption for sensor is:

$$E_{total}(d) = (E_c \cdot b) + (kd^2 \cdot b)$$

1.1. Compute the system lifetime with the sink positioned at (20, 20)

The energy consumption of each sensor and its corresponding lifetime is provided in the table 1.1.

The system's lifetime is determined by the first sensor whose battery runs out. The calculated working time for the system is 34 mins.

Table 1.1: Parking Lot

NO.	Coordinates (m)	Energy Consumption per Cycle (nJ)	Sensor Lifetime (mins)
01	(1, 2)	1470	34.01
02	(10, 3)	878	56.95
03	(4, 8)	900	55.56
04	(15, 7)	488	102.46
05	(6, 1)	1214	41.19
06	(9, 12)	470	106.38
07	(14, 4)	684	73.10
08	(3, 10)	878	56.95
09	(7, 7)	776	64.43
10	(12, 14)	300	166.67

1.2. Find the optimal sink position that maximizes system lifetime

In this task, the goal is to find the optimal sink position that maximizes the system's lifetime. By calculating the energy consumption and working time of each sensor, we determine the sensor that depletes its battery first, which defines the overall system lifetime. Using a brute-force approach, we calculated the lifetimes of all sensors for every possible sink position (from (0,0) to (20,20), all integer coordinates). We then selected the position that maximized the lifetime of the worst-case sensor. As shown in Figure 1.2, the optimal sink position is (7,8), which results in a system lifetime of 204.92 minutes.

In the 21x21 grid, the coordinate (7, 8) is near the geometric center. The geometric center is often the position that minimizes the average distance from all nodes to the sink. Placing the sink at (7, 8) helps balance the communication distances among sensors, reducing excessive energy consumption in some nodes. Furthermore, this position also balances relay task distribution, preventing certain nodes from draining their energy too quickly. Ultimately, this position achieves an optimal trade-off between transmission distance, load balancing, and individual sensor energy consumption, thus maximizing the system's overall lifetime.

1.3. Discuss the trade-offs between fixed and dynamic sink positioning

A Fixed Sink Node is placed at a fixed location during network deployment. All sensors send data to this node. The energy consumption of each sensor depends on its distance from the sink node.

Advantages and Disadvantages: Advantages: Simple design, low cost, predictable energy consumption.

Disadvantages: Sensors farther from the sink consume more energy, depleting their batteries faster and shortening the system's lifetime. System Lifetime

The system lifetime is determined by the battery depletion of the farthest sensor. Sensors farther away consume more energy, reducing the overall system lifespan.

2. Dynamic Sink Node Overview: A Dynamic Sink Node can adjust its position based on the sensor distribution, reducing transmission distance and energy consumption.

Advantages and Disadvantages:

Advantages: Balanced energy consumption, extended system lifetime, suitable for uneven

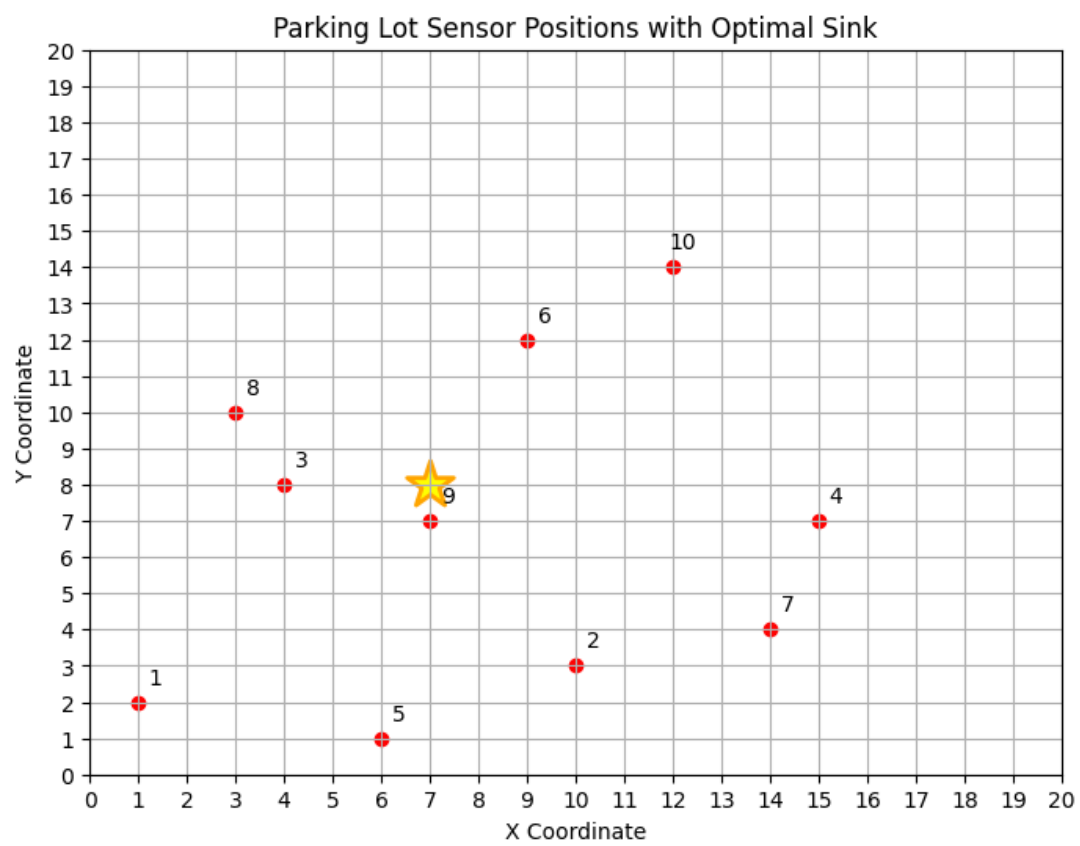


Figure 1.2: Parking Lot Sensor Positions with Optimal Sink

sensor distribution.

Disadvantages: Increased system complexity, potential latency, requires additional hardware support.

System Lifetime: By dynamically adjusting the position of the sink node, energy consumption is balanced, extending the overall system lifetime.

3. Trade-off analysis:

Fixed Sink Node: Suitable for simple scenarios with uniform sensor distribution, low cost, but shorter system lifetime. Dynamic Sink Node: Ideal for scenarios with uneven sensor distribution and higher demands for energy efficiency and system lifetime, but more complex and costly.

4. Summary:

Fixed Sink Nodes are suitable for simple applications, while Dynamic Sink Nodes are better for complex scenarios requiring higher energy efficiency and longer system lifetimes.